

Prompted Segmentation for Drywall QA

Text-conditioned segmentation pipeline that takes an image and a natural-language prompt and returns a binary mask for construction defect detection.

Supported Prompts

- `segment crack` → crack segmentation model
- `segment taping area` → drywall taping/joint segmentation model

Methodology

- Data audit** — Cracks dataset has real polygon masks; drywall dataset is box-only
- Pseudo-mask generation** — SAM with bounding box prompts used to create training masks for drywall
- Model training** — ResNet34 UNet (pretrained encoder) trained separately for each task
- Inference** — Text prompt routes to the correct model, outputs binary PNG mask

Data Preparation

Cracks dataset

- Contains real **polygon segmentation masks** for all splits (train/valid/test)
- Used directly for supervised training — no additional preparation needed

Drywall dataset

- Labels are **bounding boxes only** — no pixel masks in the annotations
- SAM (Segment Anything Model)** was used to generate pseudo-masks:
 - Each bounding box was passed as a prompt to SAM
 - The predicted mask was hard-clipped to the bbox boundary to prevent leakage
 - All instance masks per image were merged into one binary mask
- v2 pseudo-masks (no bbox padding + hard clipping) were used as the final training targets
- Stricter post-processing versions were tested but discarded — they over-filtered and left most masks empty

Results

Prompt	Dice	mIoU	GT type
<code>segment crack</code>	0.7045	0.5438	True polygon GT
<code>segment taping area</code>	0.7070	0.5468	SAM pseudo-GT

Run Inference



```
pip install segmentation-models-pytorch torch torchvision

python phase6_inference.py /path/to/project \
    --image_dir /path/to/images \
    --prompt "segment crack" \
    --output_dir /path/to/outputs
```